

Exercise 2: Implementing The Factory Method Pattern

CODE:

```
interface Document {
    void open();
    void save();
    void close();
}

class WordDocument implements Document {
    @Override
    public void open() {
        System.out.println("Opening Word Document (.docx)...");
    }
    @Override
    public void save() {
        System.out.println("Saving changes to Word Document.");
    }
    @Override
    public void close() {
        System.out.println("Closing Word Document.");
    }
}

class PdfDocument implements Document {
    @Override
    public void open() {
        System.out.println("Opening PDF Document (.pdf) in reader mode...");
    }
    @Override
    public void save() {
        System.out.println("Saving annotations to PDF Document.");
    }
    @Override
    public void close() {
        System.out.println("Closing PDF Document.");
    }
}

class ExcelDocument implements Document {
    @Override
    public void open() {
        System.out.println("Opening Excel Spreadsheet (.xlsx)...");
    }
}
```

```

        @Override
        public void save() {
            System.out.println("Recalculating formulas and saving Excel
Spreadsheet.");
        }
        @Override
        public void close() {
            System.out.println("Closing Excel Spreadsheet.");
        }
    }

    abstract class DocumentFactory {
        public abstract Document createDocument();
        public void manageDocument() {
            Document doc = createDocument();
            doc.open();
            doc.save();
            doc.close();
        }
    }

    class WordDocumentFactory extends DocumentFactory {
        @Override
        public Document createDocument() {
            return new WordDocument();
        }
    }

    class PdfDocumentFactory extends DocumentFactory {
        @Override
        public Document createDocument() {
            return new PdfDocument();
        }
    }

    class ExcelDocumentFactory extends DocumentFactory {
        @Override
        public Document createDocument() {
            return new ExcelDocument();
        }
    }

    public class FactoryMethodPatternDemo {
        public static void main(String[] args) {
            System.out.println("=== Factory Method Pattern Implementation Demo ===");
        }
    }

```

```
DocumentFactory wordFactory = new WordDocumentFactory();
DocumentFactory pdfFactory = new PdfDocumentFactory();
DocumentFactory excelFactory = new ExcelDocumentFactory();

System.out.println("\n--- Processing Word Document ---");
Document wordDoc = wordFactory.createDocument();
wordDoc.open();
wordDoc.save();
wordDoc.close();

System.out.println("\n--- Processing PDF Document ---");
Document pdfDoc = pdfFactory.createDocument();
pdfDoc.open();
pdfDoc.save();
pdfDoc.close();

System.out.println("\n--- Processing Excel Document ---");
Document excelDoc = excelFactory.createDocument();
excelDoc.open();
excelDoc.save();
excelDoc.close();

System.out.println("\n=====");
System.out.println("SUCCESS: Factory Method pattern working correctly!");
System.out.println("=====");
    }
}
```

OUTPUT:

```
● PS D:\Github Projects\Cognizant-Hands-On-Exercise\Module 1\Exercise 2> cd "d:\Github Projects\Cognizant-Hands-On-Exercise\Module 1\Exercise 2\" ; if ($?) { javac FactoryMethodPatternDemo.java } ; if ($?) { java FactoryMethodPatternDemo }
=== Factory Method Pattern Implementation Demo ===

--- Processing Word Document ---
Opening Word Document (.docx)...
Saving changes to Word Document.
Closing Word Document.

--- Processing PDF Document ---
Opening PDF Document (.pdf) in reader mode...
Saving annotations to PDF Document.
Closing PDF Document.

--- Processing Excel Document ---
Opening Excel Spreadsheet (.xlsx)...
Recalculating formulas and saving Excel Spreadsheet.
Closing Excel Spreadsheet.

=====
SUCCESS: Factory Method pattern working correctly!
=====
○ PS D:\Github Projects\Cognizant-Hands-On-Exercise\Module 1\Exercise 2> █
```